

Tristan Toshiharu Endo

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Education

University of California, San DiegoSept 2024 – Jun 2027
(Expected)

B.S. Math-Computer Science

- Relevant Coursework: Real Analysis, Stochastic Processes, Brownian Motion, Mathematics in Finance, Enumerative Combinatorics, Probability Theory, Abstract Algebra

Research Experience

Undergraduate Researcher — Advisor: Alexander CloningerJan 2026 – Mar 2026

Neural Network Gaussian Processes & Conjugate Kernels

- Derived closed-form conjugate kernels for infinite-width ReLU and erf networks, confirming alignment between finite-width simulations and theoretical kernel recursion predictions
- Investigated kernel degeneracy, spectral bias, and expressivity as functions of activation choice and weight variance, comparing finite-width networks to their infinite-width Gaussian process limits

Undergraduate Researcher — Advisor: Raymond O. ChouFebruary 2026 – Present

Springer Fibers (Algebraic Geometry & Combinatorics)

- Investigate the geometry and combinatorics of Springer fibers, analyzing flag variety structure and symmetric group actions to characterize irreducible components

Industry Experience

Quantitative Researcher / AI Engineering InternJanuary 2026 – Present

Tekly Studio — Quantitative Research & AI Engineering LabSan Francisco, CA

- Developed and backtested ML-driven trading strategies on financial time series, evaluating performance via Sharpe ratio and drawdown analysis
- Engineered data pipelines for preprocessing, simulation, and model evaluation, enabling rapid iteration across strategy variants

Audio Engineer / AnalystApril 2025 – Present

Think Twice LegalSan Bruno, CA

- Optimized Demucs-based source separation models via hyperparameter tuning, improving reconstruction fidelity for forensic audio analysis in legal proceedings
- Built preprocessing and normalization pipelines for consistent model input across varying sample rates and noise conditions

Selected Projects

Stochastic Simulation of Brownian Motion & OU Processes

- Simulated fractional Brownian motion across varying Hurst parameters to characterize long-range dependence and correlation structure
- Quantified empirical distributions of extrema via large-scale Monte Carlo simulation, validating convergence to theoretical predictions across parameter regimes

Audio Signal Processing & ML Pipeline Development

- Designed FFT-based noise reduction models to isolate target signals from complex audio environments
- Built automated stem separation pipelines using Demucs with normalization workflows for consistent model input across varied audio conditions

Technologies

Languages: C, C++, Java, Matlab, Python, R, SQL

Libraries & Tools: PyTorch, TensorFlow, NumPy, SciPy, Torchaudio, FFmpeg, L^AT_EX, VS Code, RStudio, Tableau, SQL Workbench

Areas: Algorithmic Trading, Time-Series Modeling, Stochastic Processes, Audio Source Separation, Statistical Modeling